

LEDGER · 2026-08-31 · Dermo-Cosmetics

Microbial Lactobionic Acid (LBA) via Whole-Cell Dairy Byproduct Fermentation

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Glycolic Acid
Measured advantage	2.5x lower barrier disruption and >2.0x reduction in erythema vs glycolic acid
Time to first revenue	6 months
Gross margin at parity	83.3%
Startup CapEx	\$202,000
Payback from first sale	19.8 months
Capital productivity	0.96x
Regulatory risk	Moderate — qualification costs reported (\$65,000 estimated)
Assessment	

LBA fails the strict superiority requirement because its head-to-head advantages focus on reduced barrier disruption and decreased erythema, which are safety and mildness preference attributes rather than performance superiority on exfoliation/cell turnover (where it only matches Glycolic Acid).

Also flagged

- Evidence rests on non-peer-reviewed sources
- Price parity asserted, not sourced
- no_production_runsheet
- unsourced_equipment
- no_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-31 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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